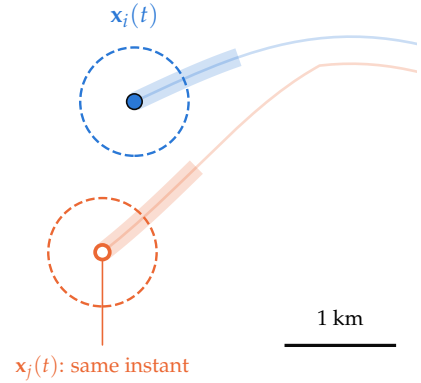
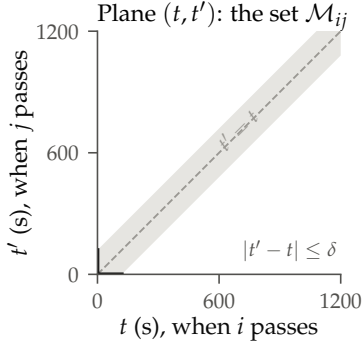
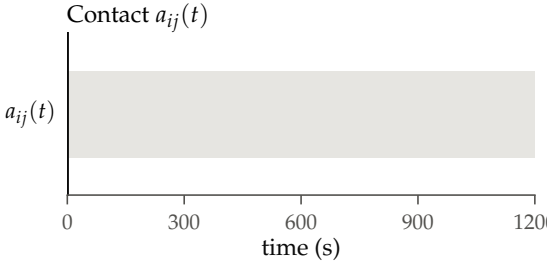
$t = 0$ s  $j$ passes by $i$: no  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



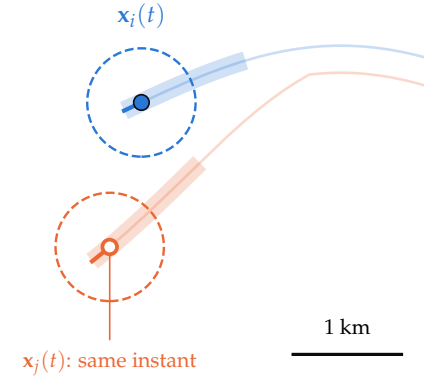



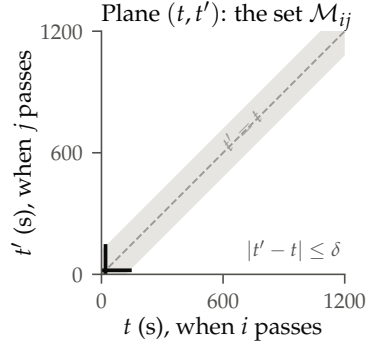



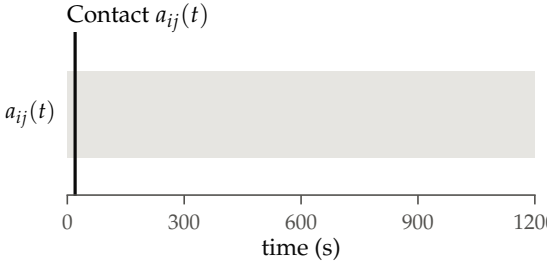
$t = 20$ s  $j$ passes by $i$: no  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



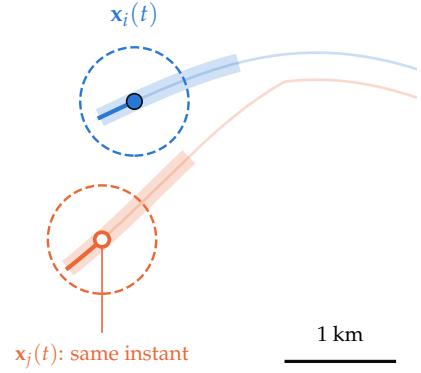



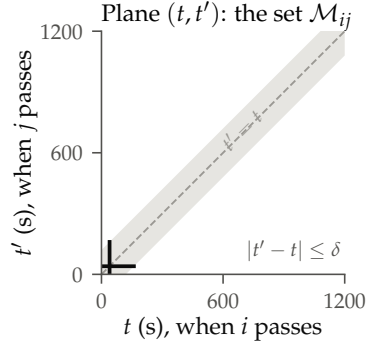



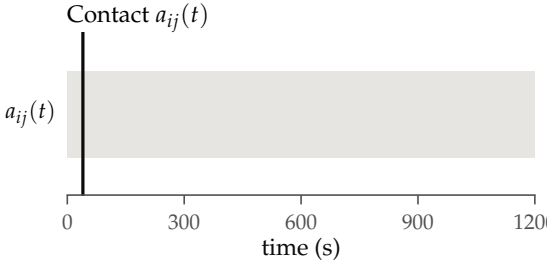
$t = 40$ s  $j$ passes by $i$: no  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



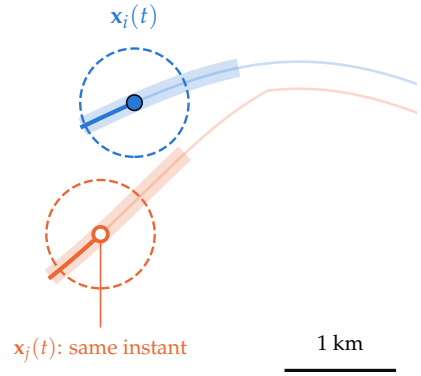



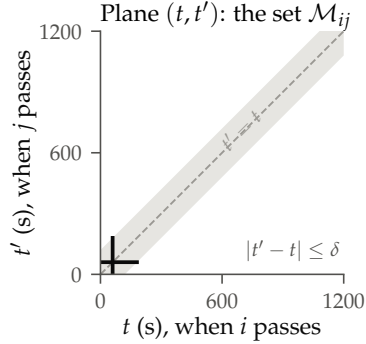



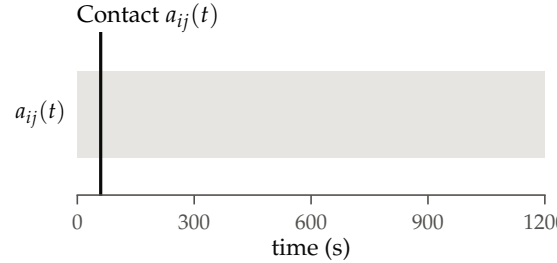
$t = 60$ s  $j$ passes by $i$: no  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



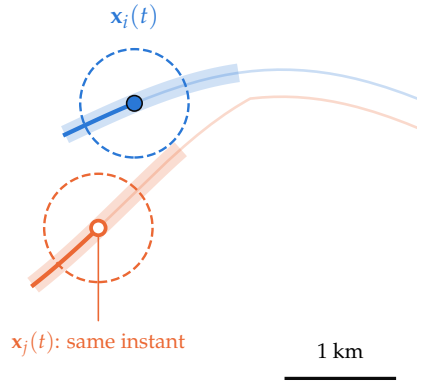



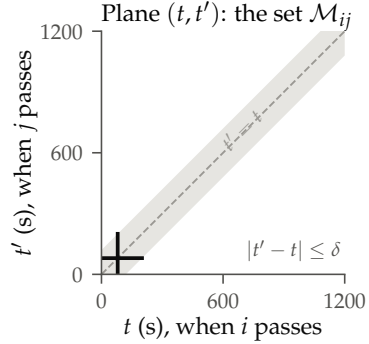



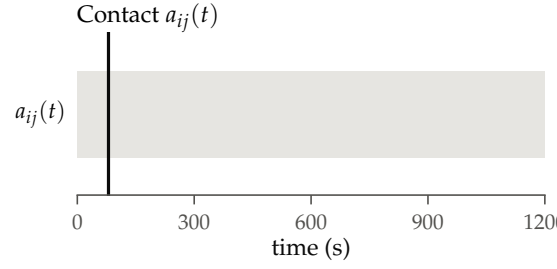
$t = 80$ s  $j$ passes by $i$: no  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



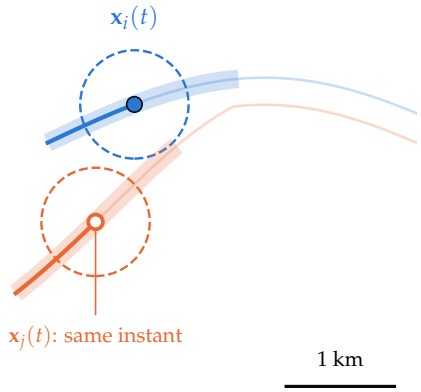



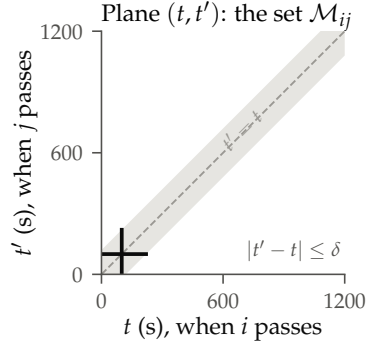



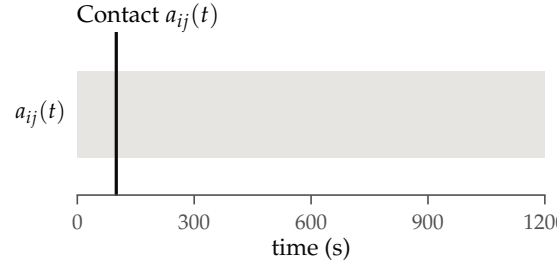
$t = 100$ s  $j$ passes by $i$: no  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



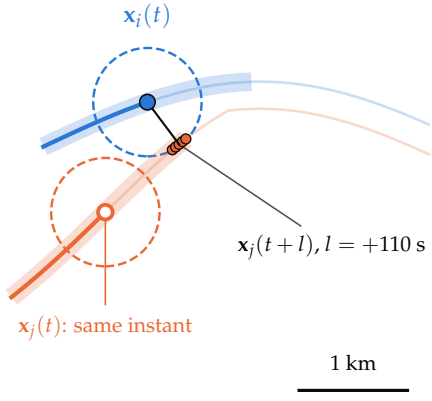



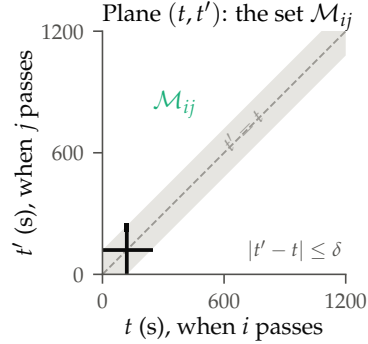



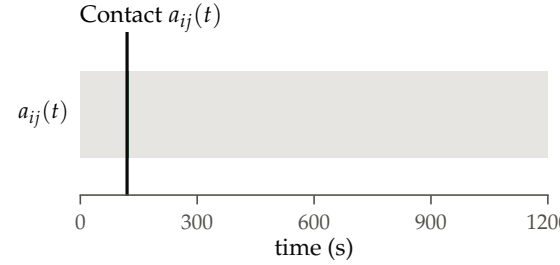
$t = 120$ s  $j$ passes by $i$: yes  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



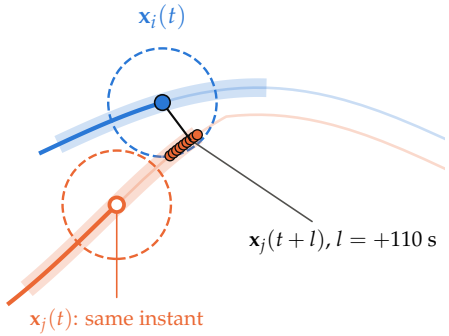



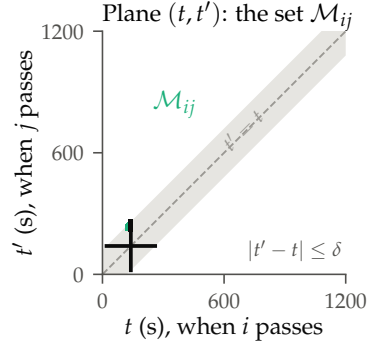



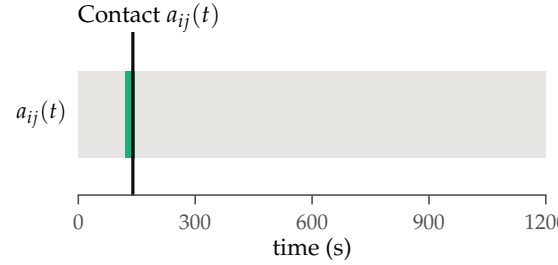
$t = 140$ s  $j$ passes by $i$: yes  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



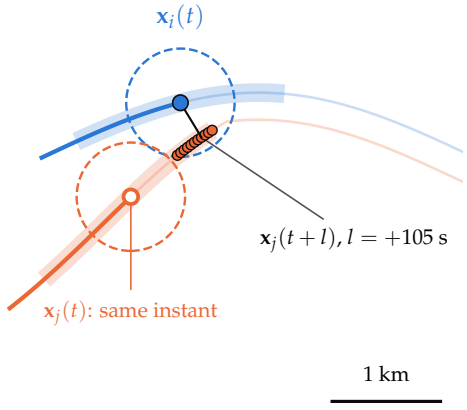



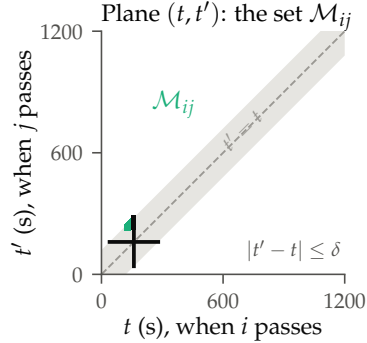



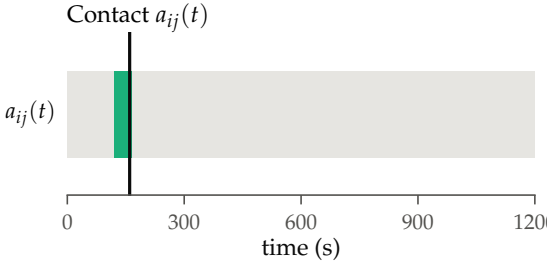
$t = 160$ s  $j$ passes by $i$: yes  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



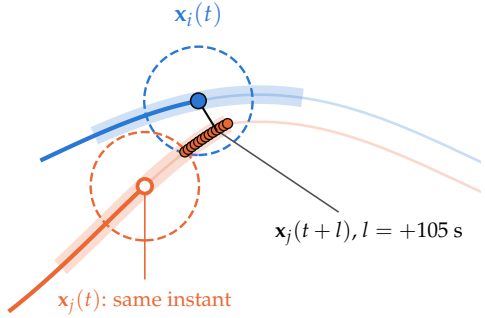



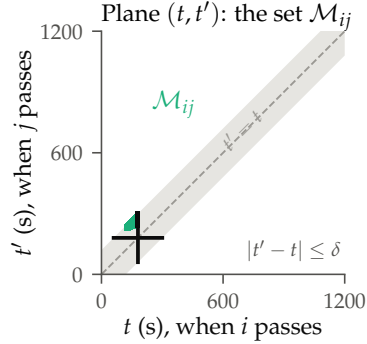



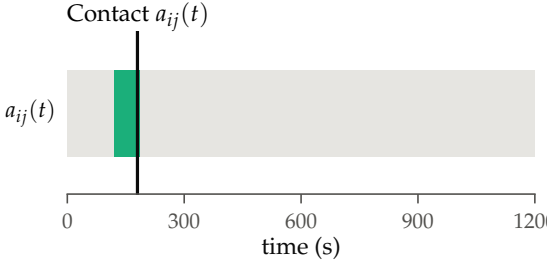
$t = 180$ s  $j$ passes by $i$: yes  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



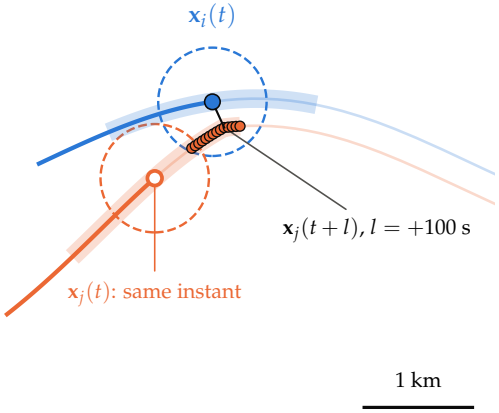



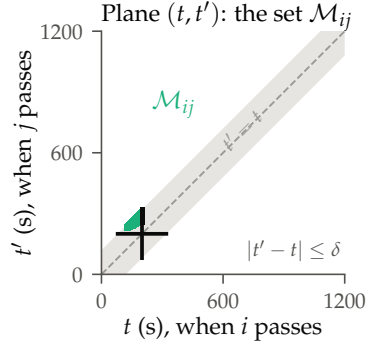



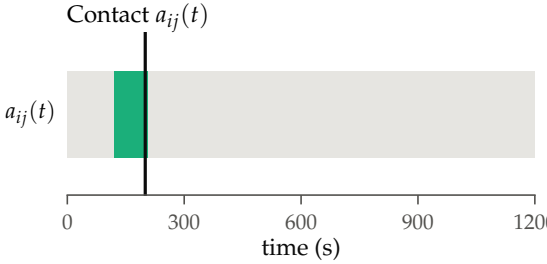
$t = 200$ s  $j$ passes by $i$: yes  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



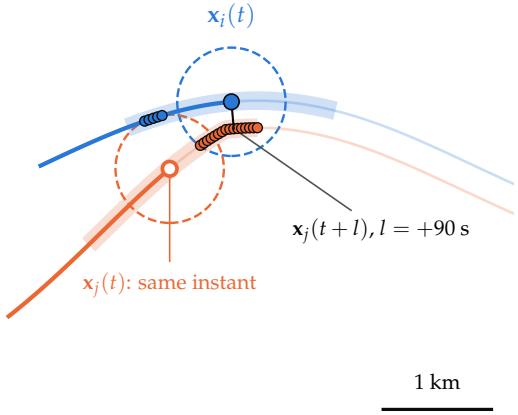



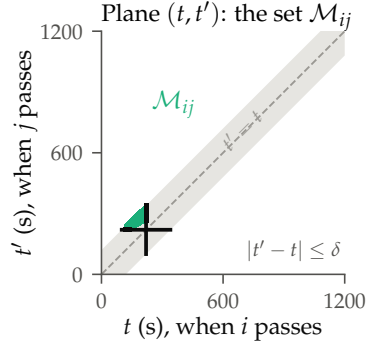



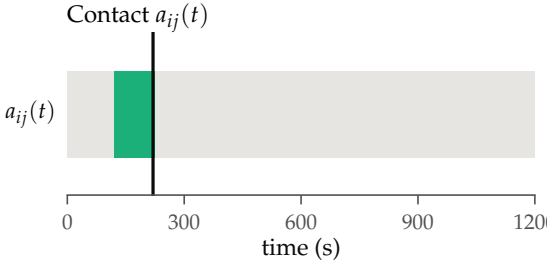
$t = 220$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



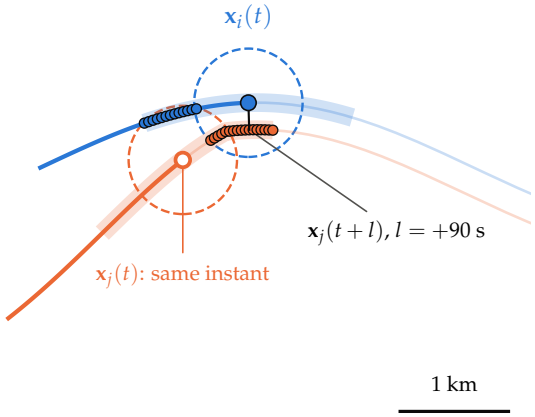



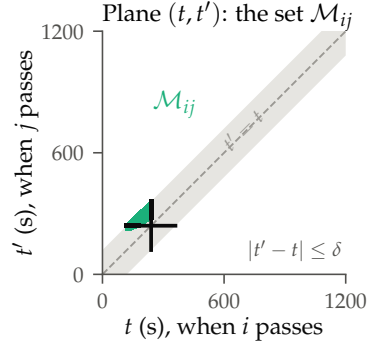



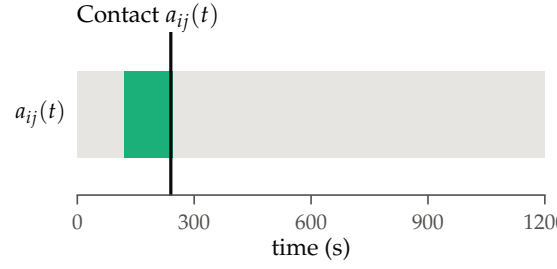
$t = 240$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



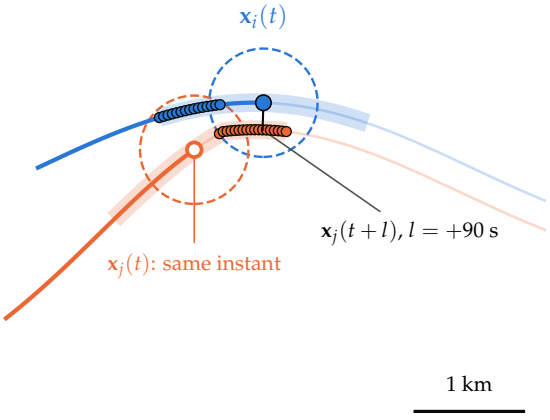



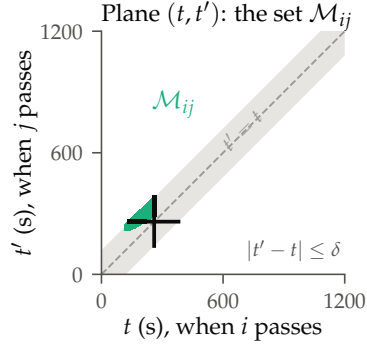



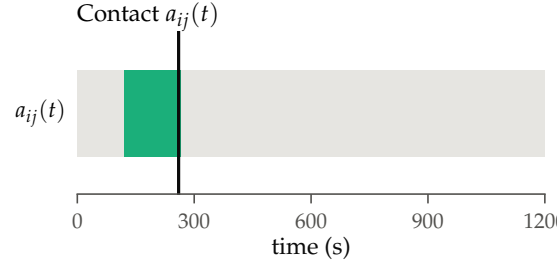
$t = 260$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



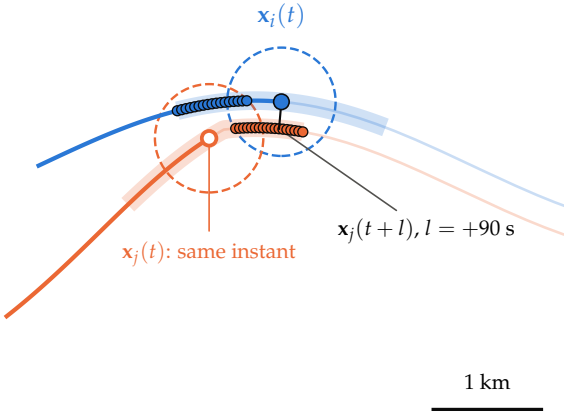



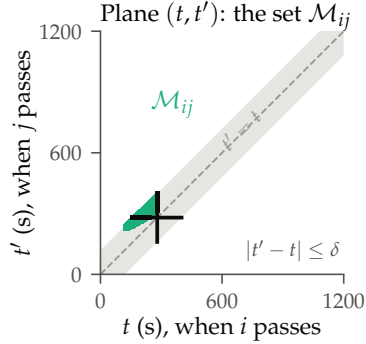



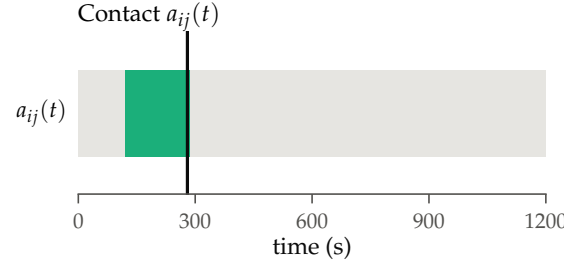
$t = 280$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



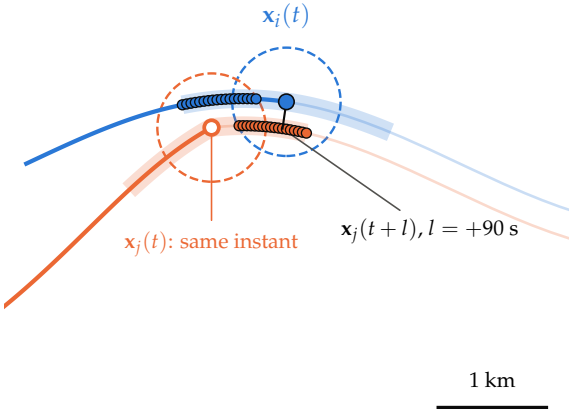



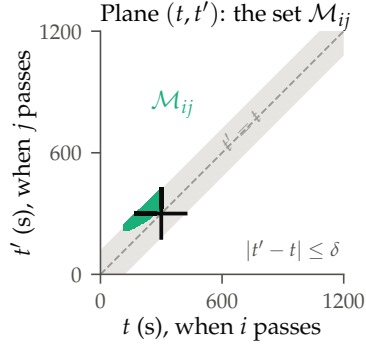



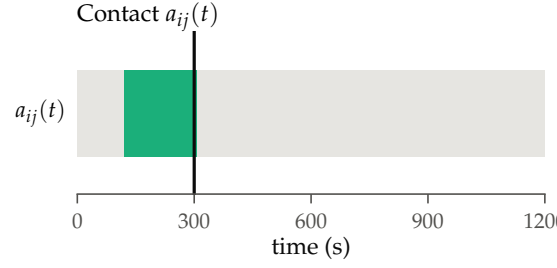
$t = 300$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



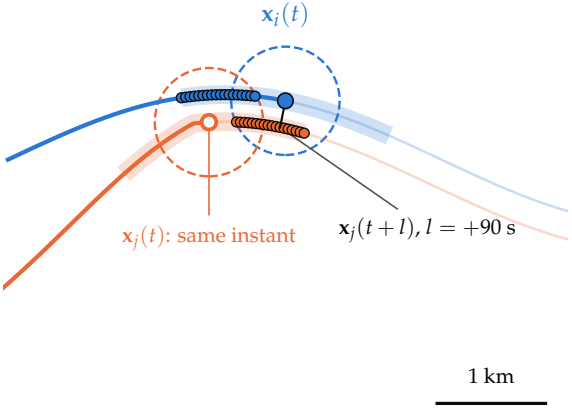



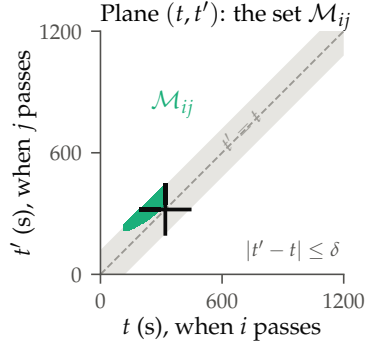



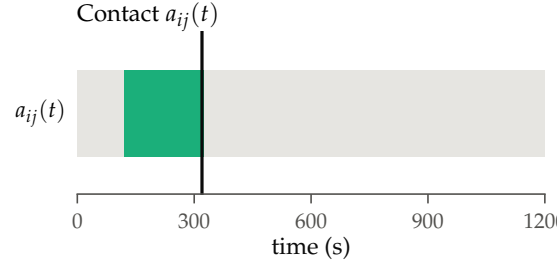
$t = 320$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



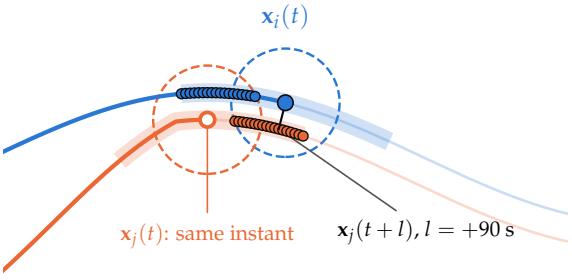



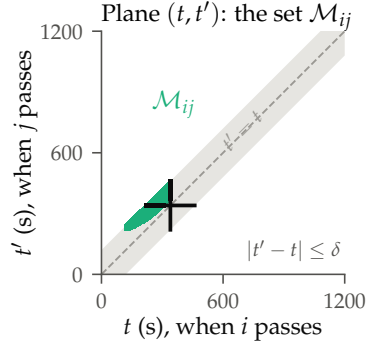



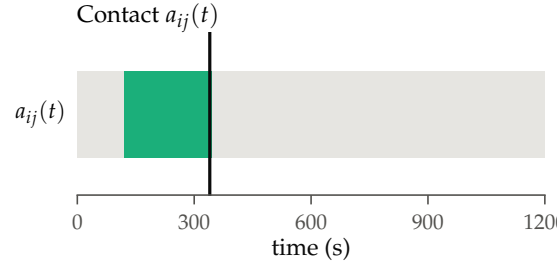
$t = 340$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



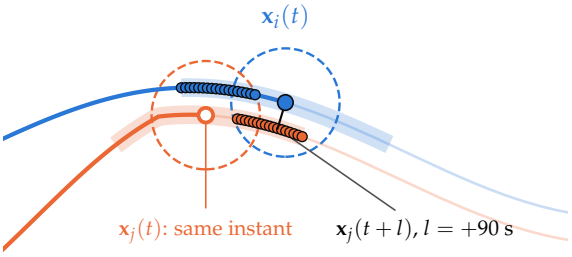



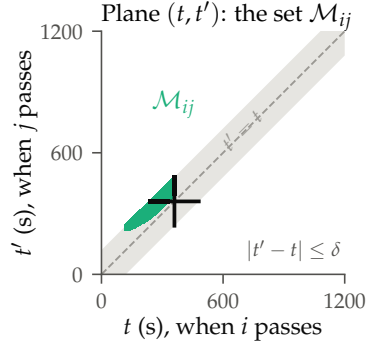



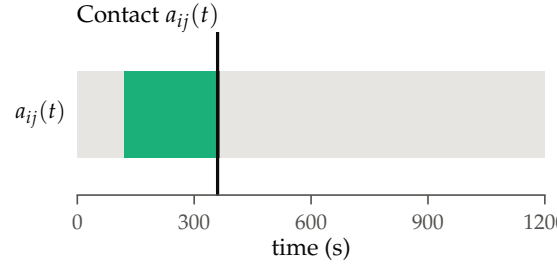
$t = 360$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



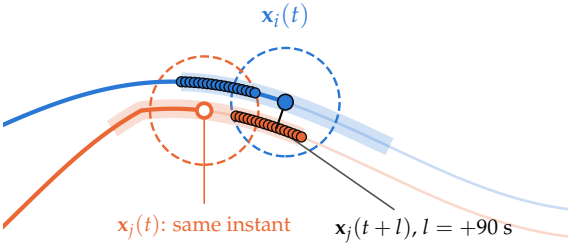



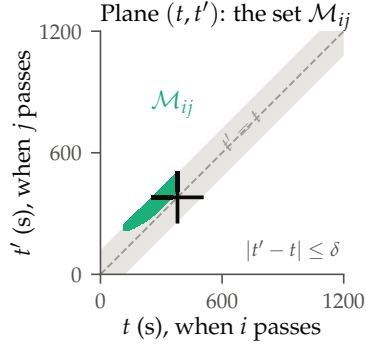
$t = 380$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

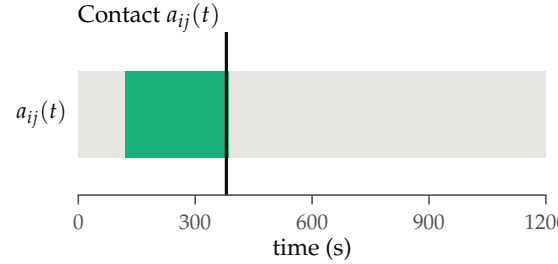
**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$




1 km

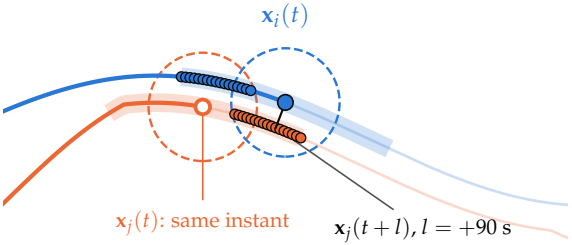


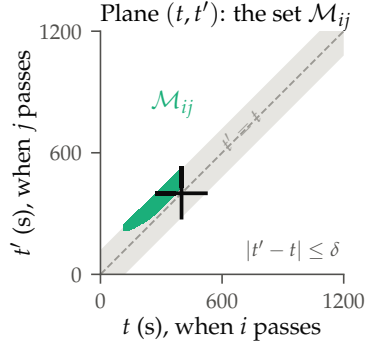
$t = 400$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

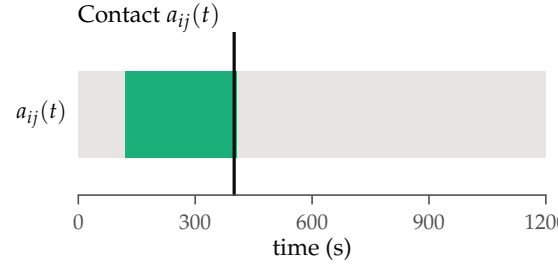
**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$




1 km

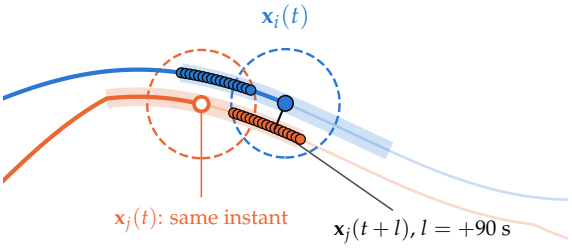


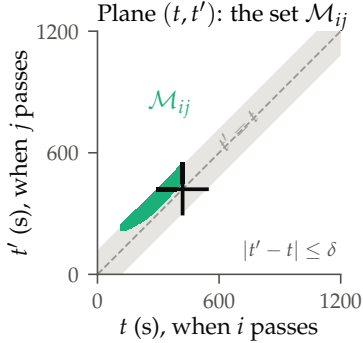



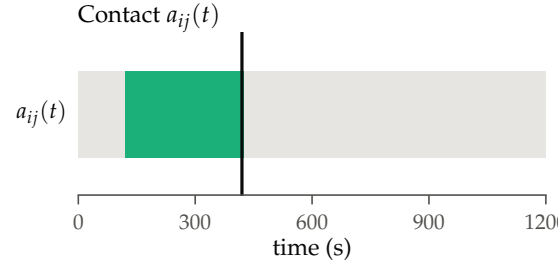
$t = 420$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



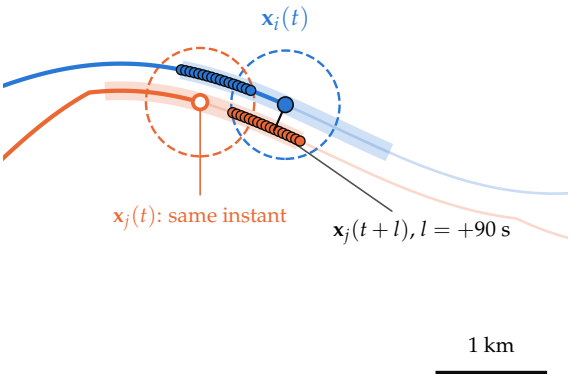



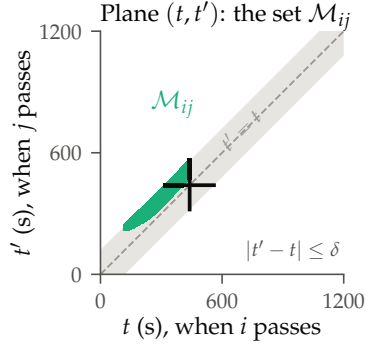



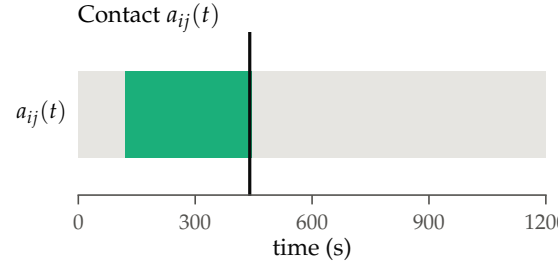
$t = 440$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



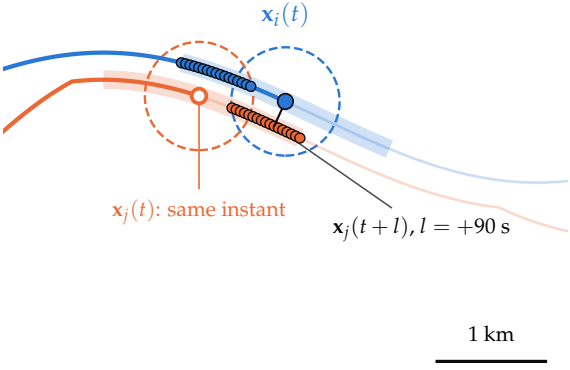



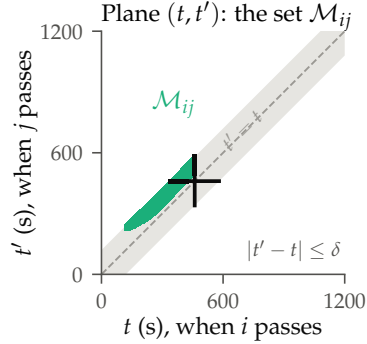



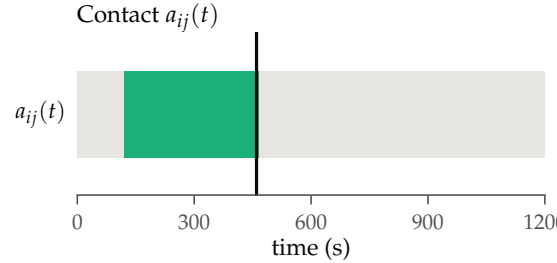
$t = 460$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



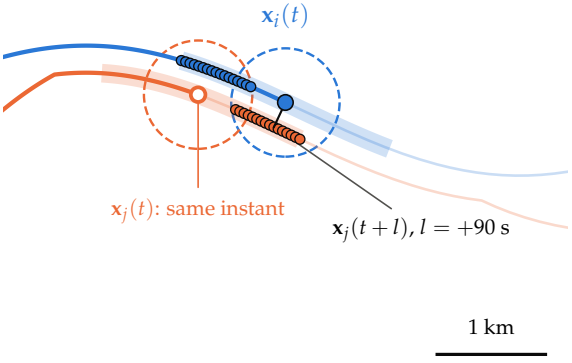



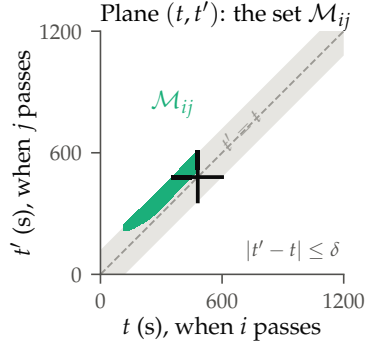



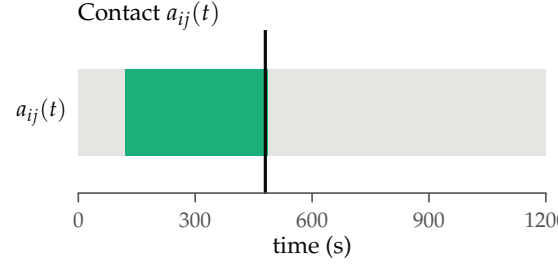
$t = 480$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



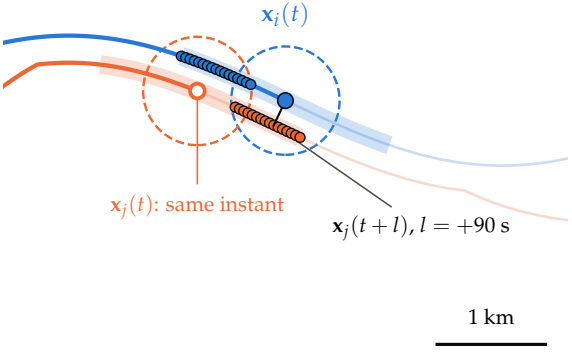



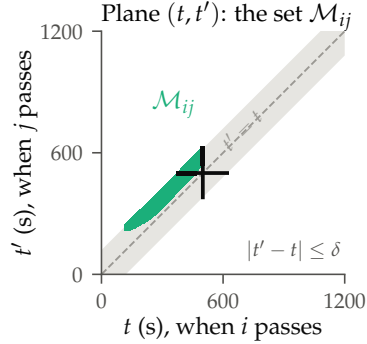



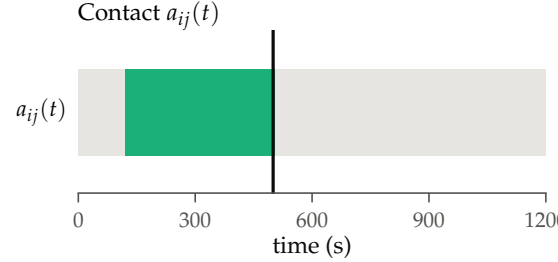
$t = 500$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



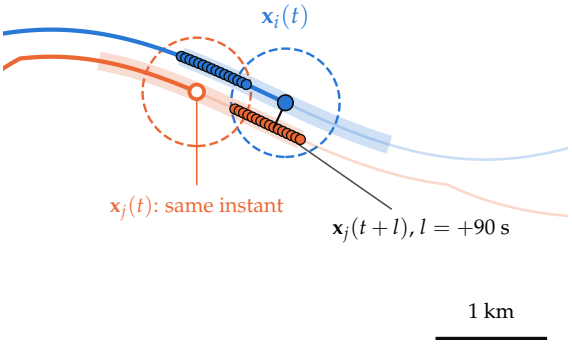



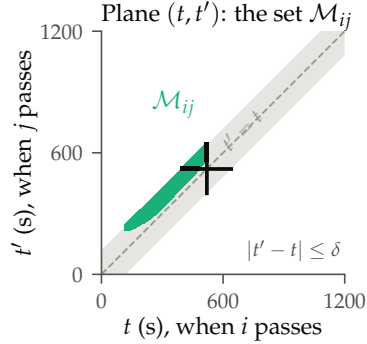



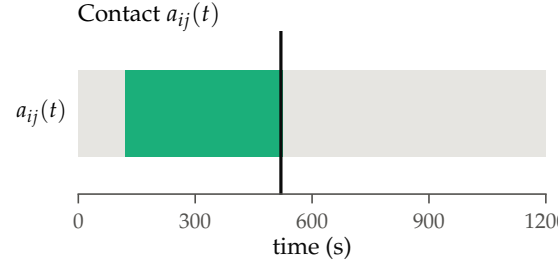
$t = 520$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



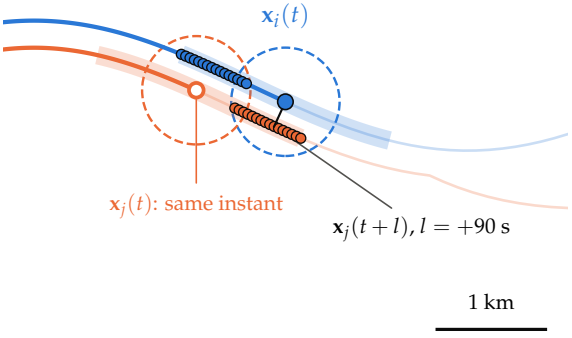



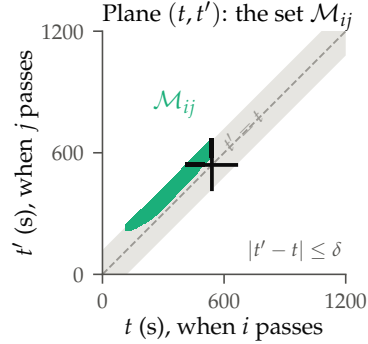



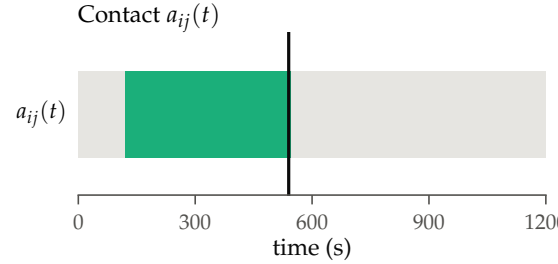
$t = 540$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



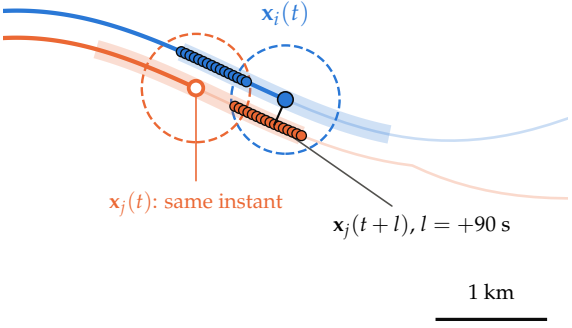



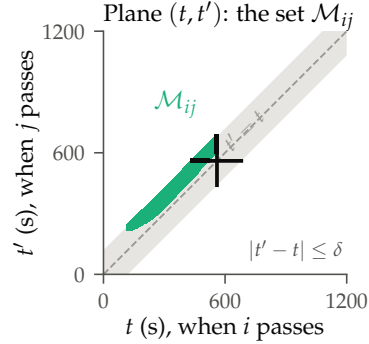



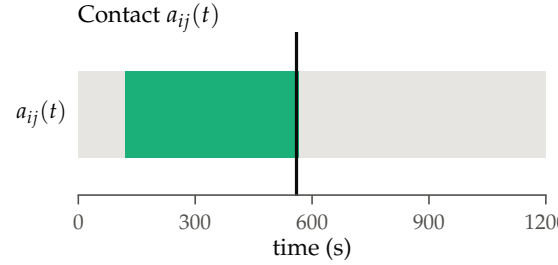
$t = 560$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



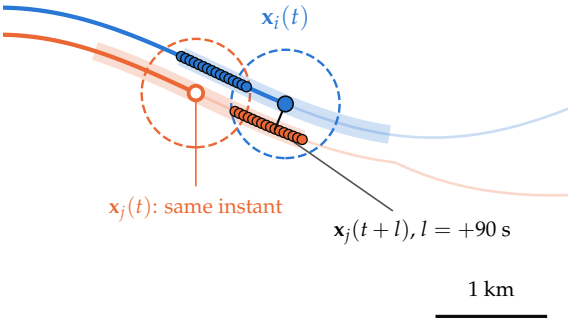



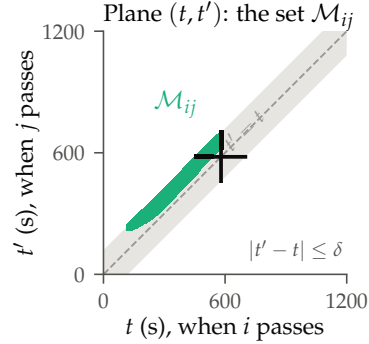



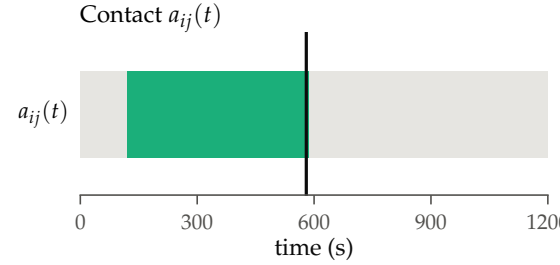
$t = 580$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



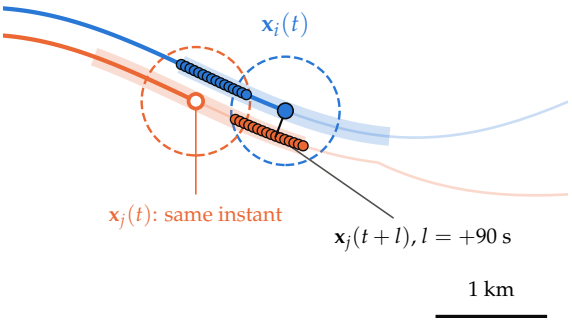



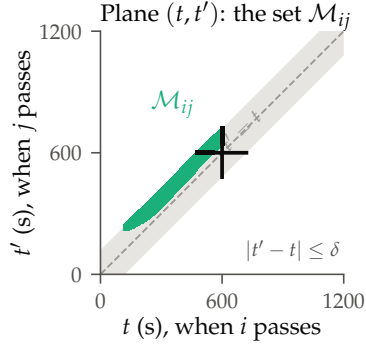



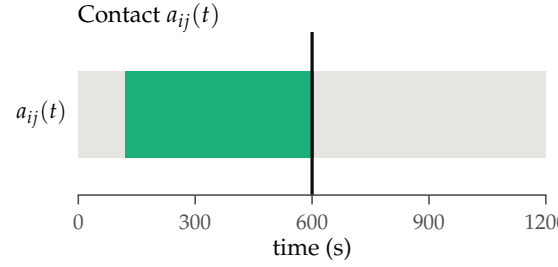
$t = 600$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



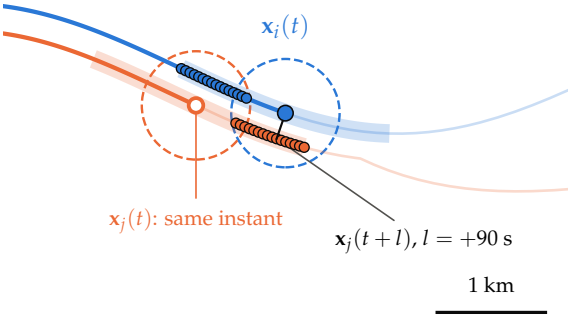



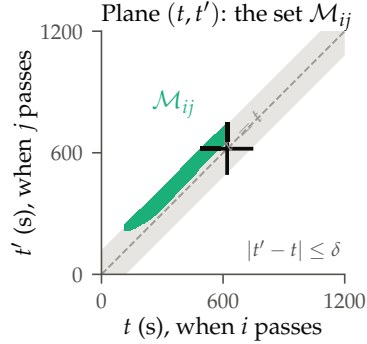



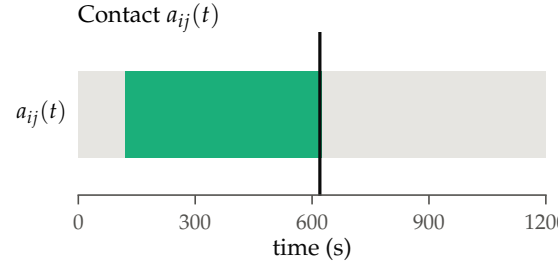
$t = 620$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



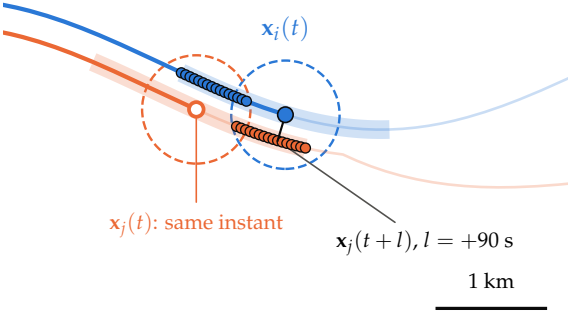



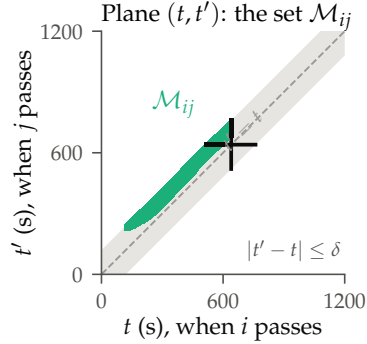



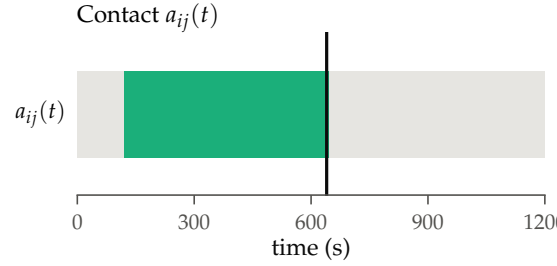
$t = 640$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



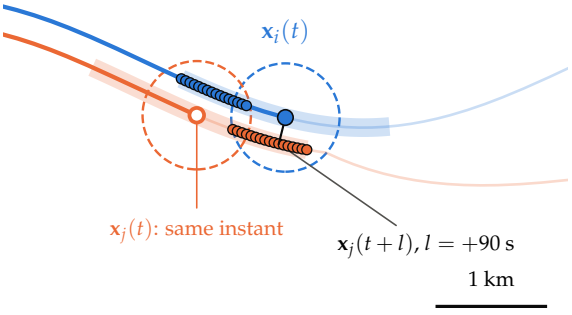



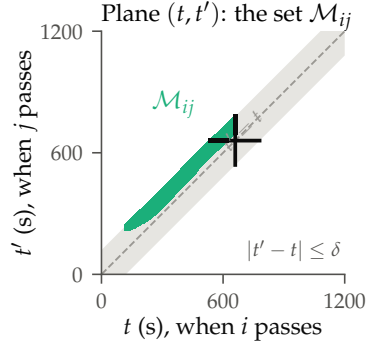



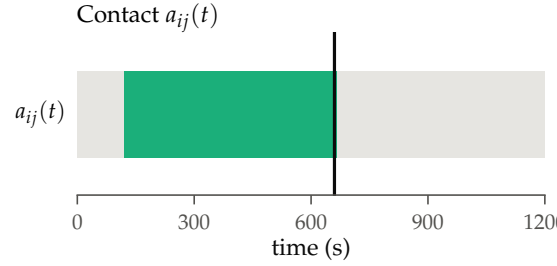
$t = 660$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



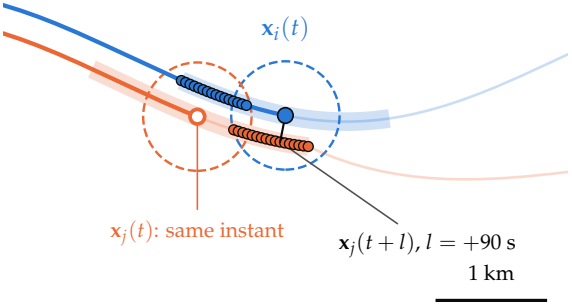



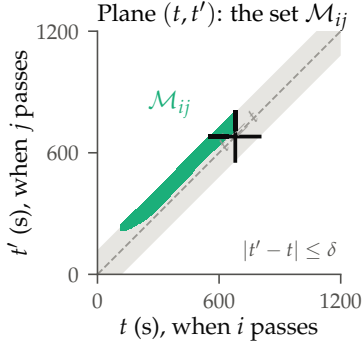



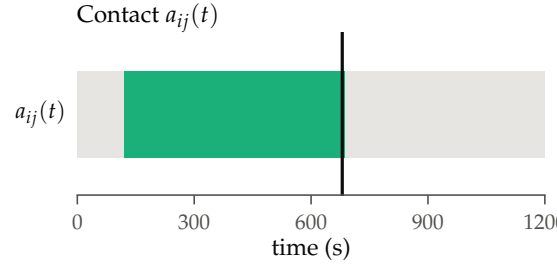
$t = 680$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



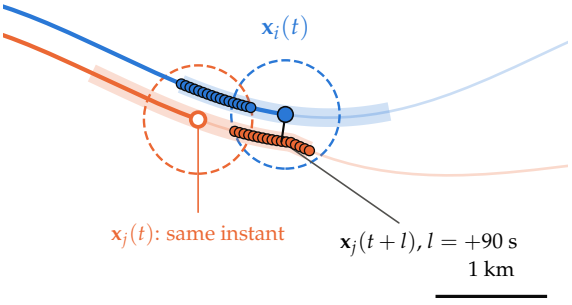



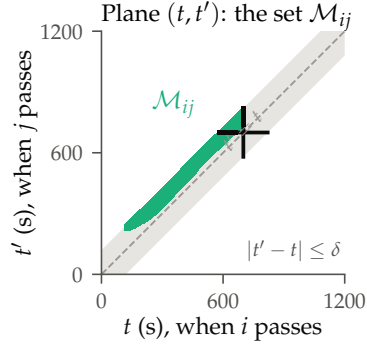



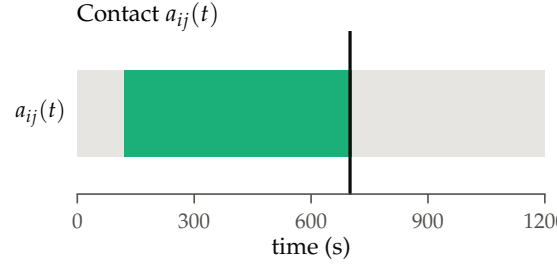
$t = 700$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



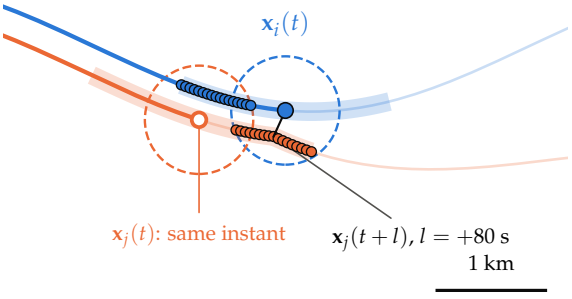



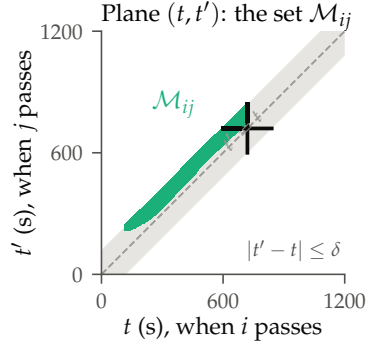



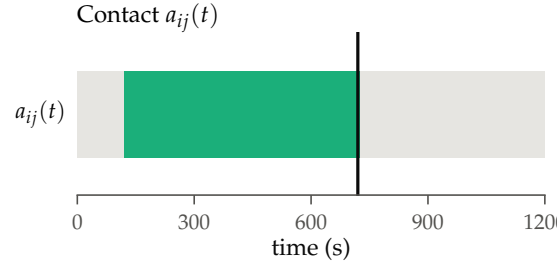
$t = 720$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



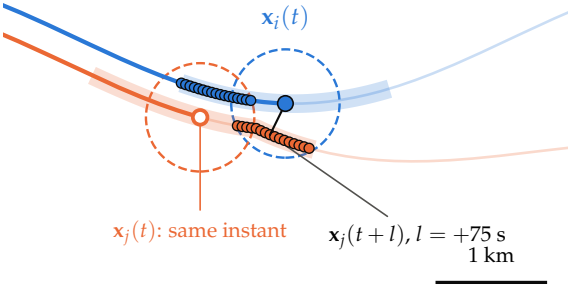



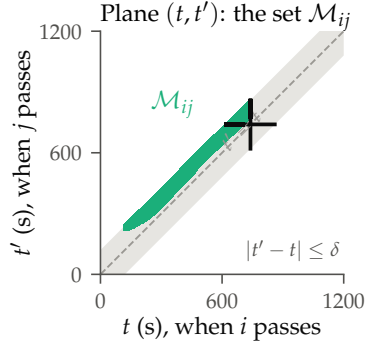



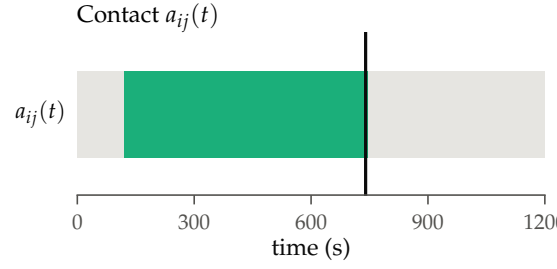
$t = 740$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



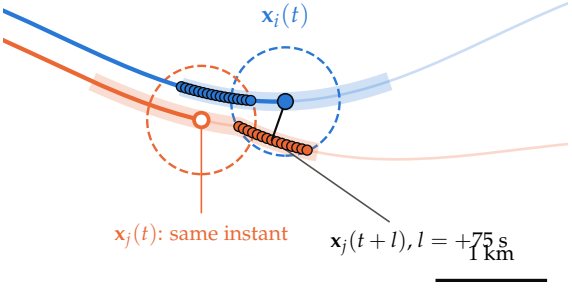



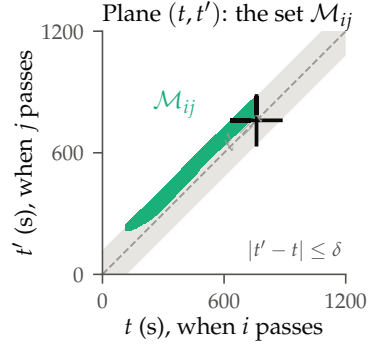



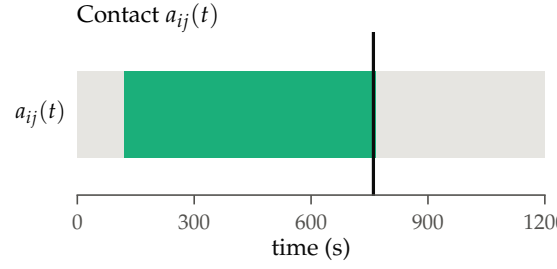
$t = 760$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



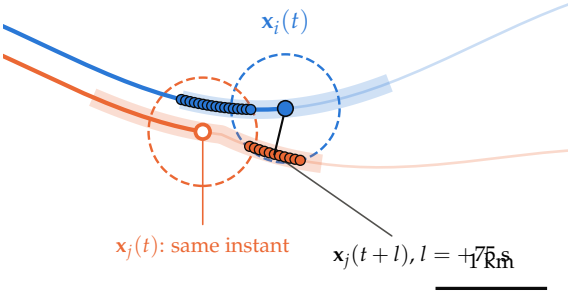



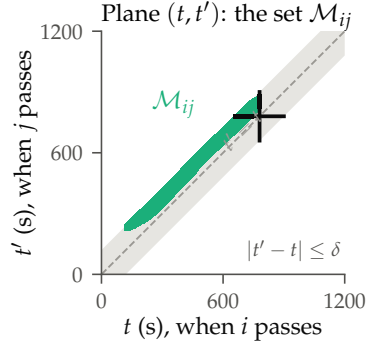



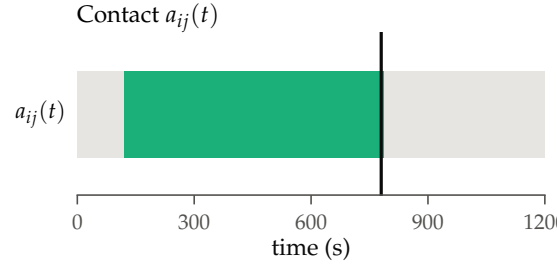
$t = 780$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



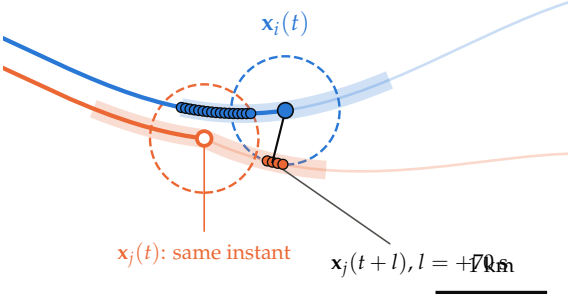



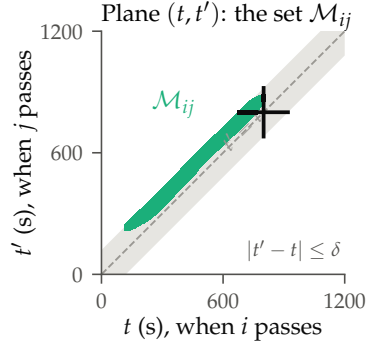



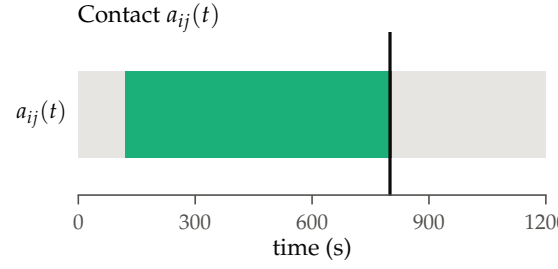
$t = 800$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



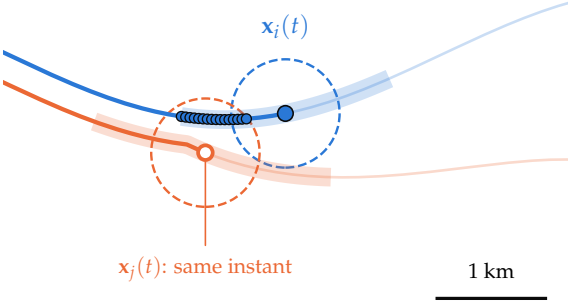



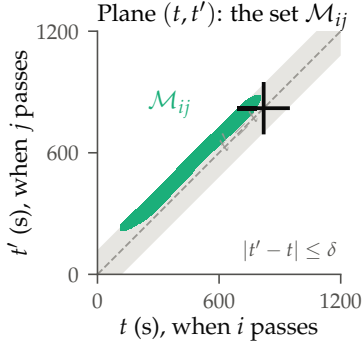



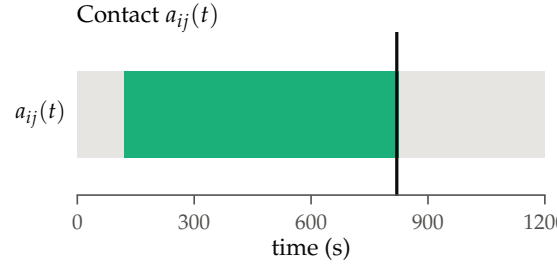
$t = 820$ s  $j$ passes by $i$: no  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



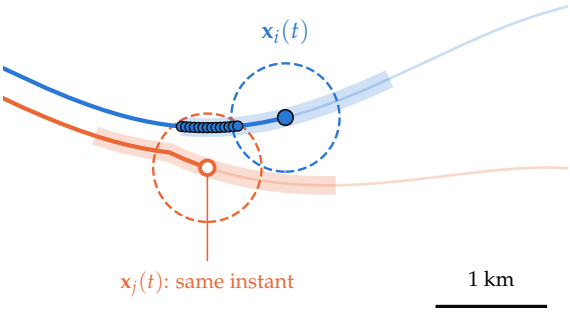



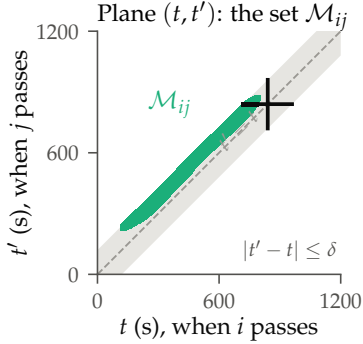



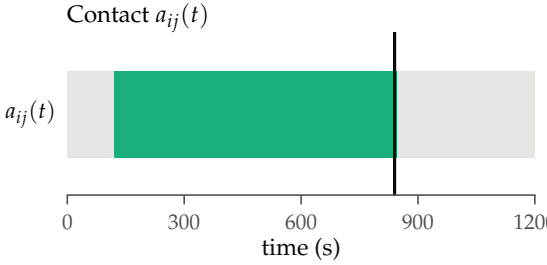
$t = 840$ s  $j$ passes by $i$: no  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



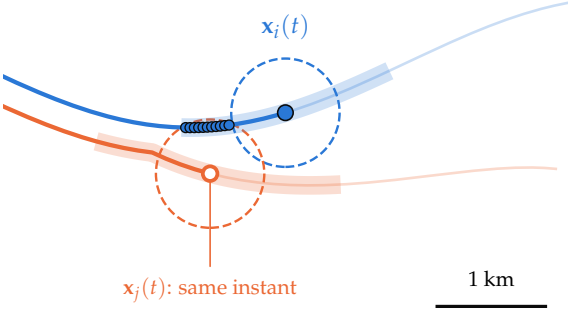



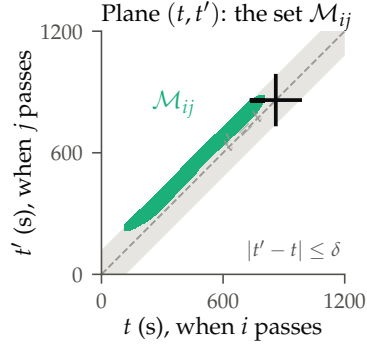



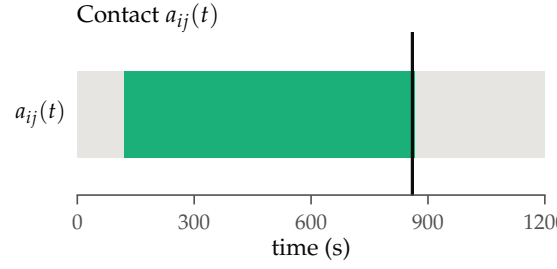
$t = 860$ s  $j$ passes by $i$: no  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



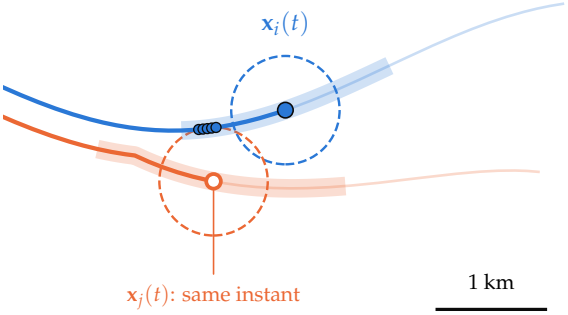



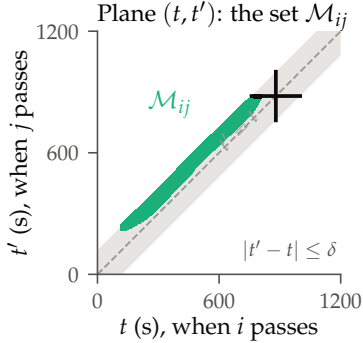



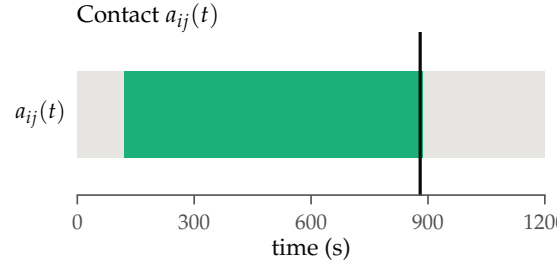
$t = 880$ s  $j$ passes by $i$: no  $i$ passes by $j$: yes

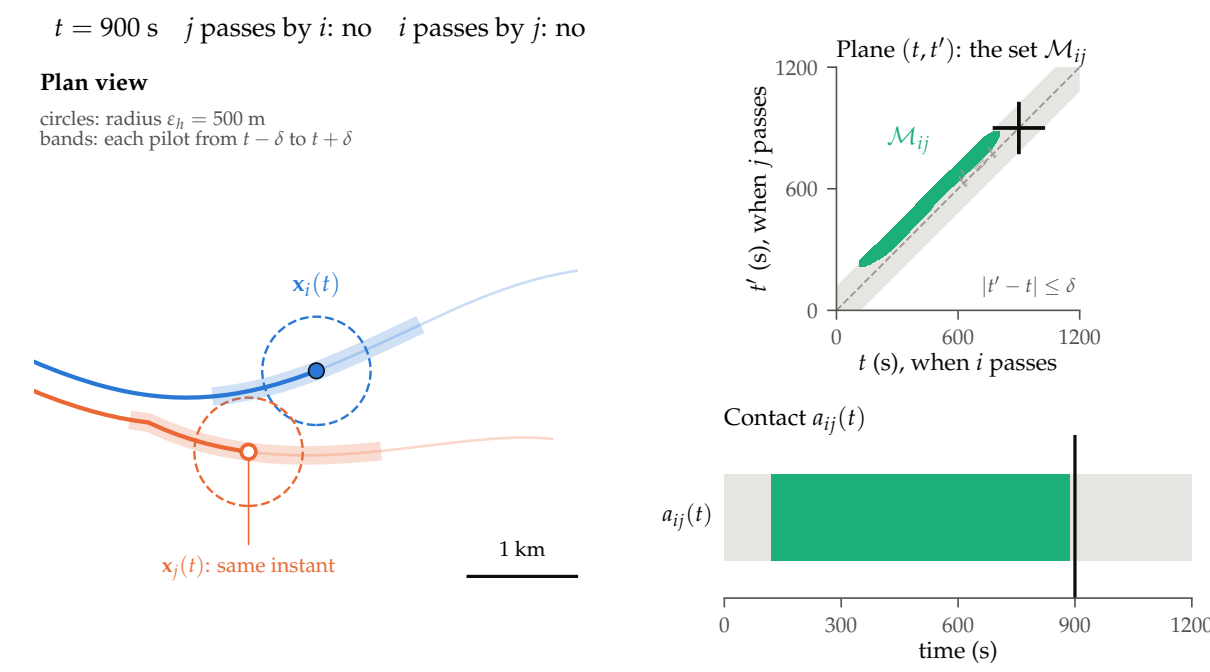
**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



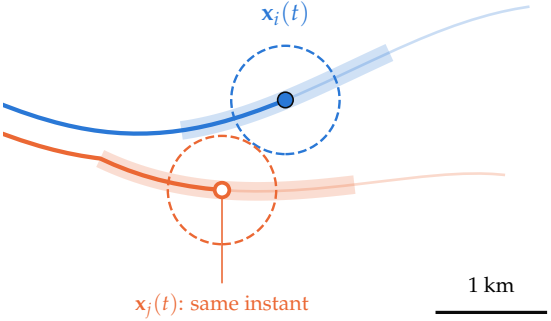



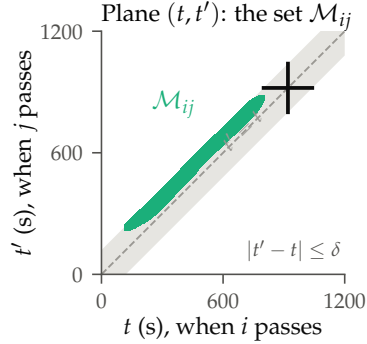

$t = 900$ s  $j$ passes by $i$: no  $i$ passes by $j$: no
Plan view
circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$
$\mathbf{x}_i(t)$
$\mathbf{x}_j(t)$: same instant
1 km
Plane $(t, t')$: the set $\mathcal{M}_{ij}$
$\mathcal{M}_{ij}$
$|t' - t| \leq \delta$
$t'$ (s), when $j$ passes
$t$ (s), when $i$ passes
1200
600
0
0
600
1200
Contact $a_{ij}(t)$
$a_{ij}(t)$
0
300
600
900
1200
time (s)

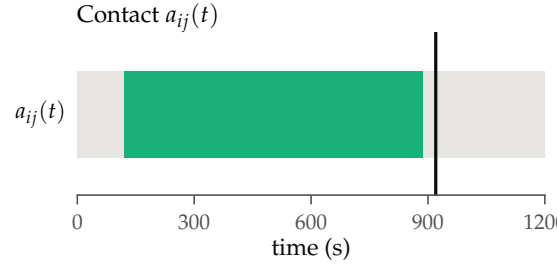
$t = 920$ s  $j$ passes by $i$: no  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



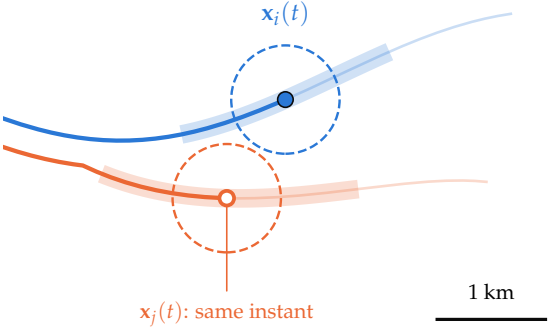



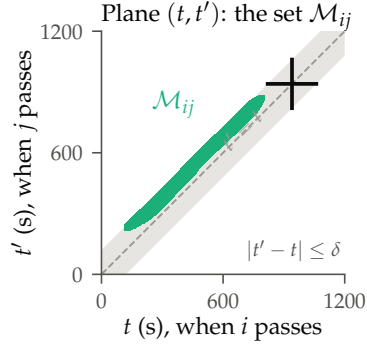



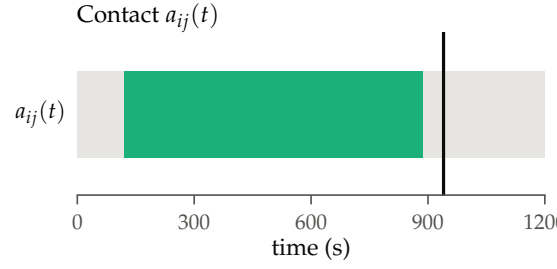
$t = 940$ s  $j$ passes by $i$: no  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



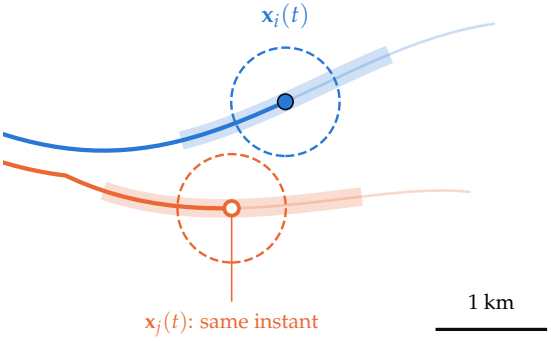



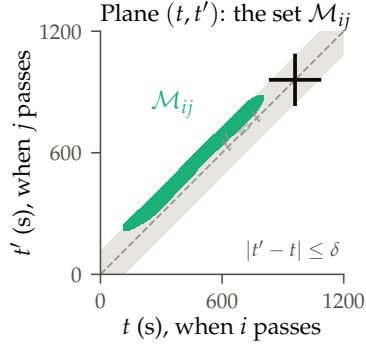



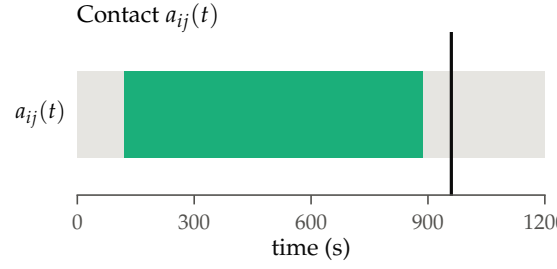
$t = 960$ s  $j$ passes by $i$: no  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



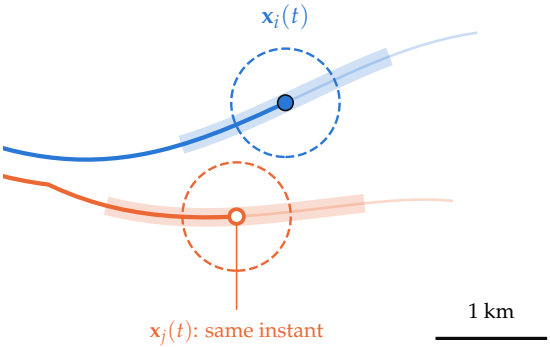



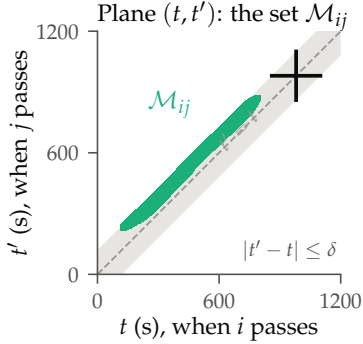



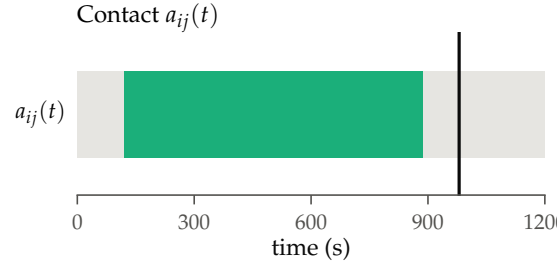
$t = 980$ s  $j$ passes by $i$: no  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



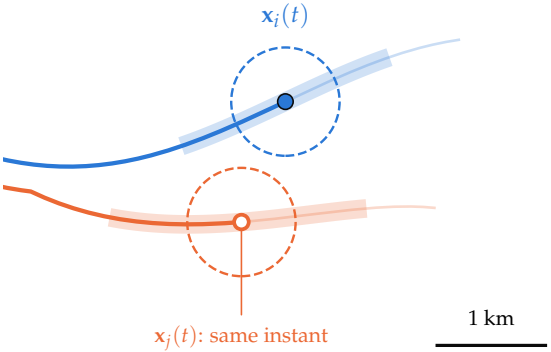



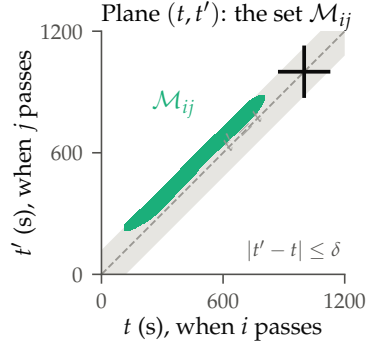



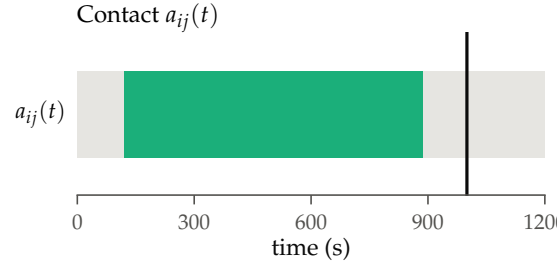
$t = 1000$ s  $j$ passes by $i$: no  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



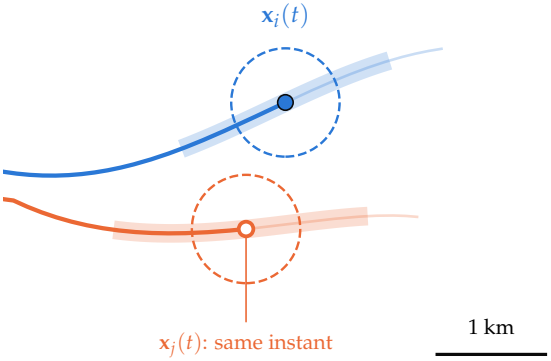



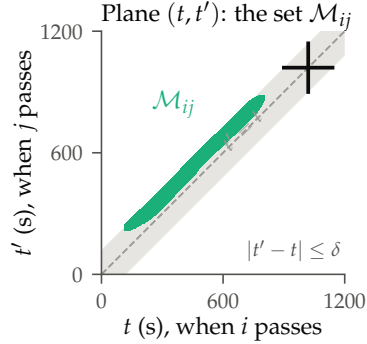



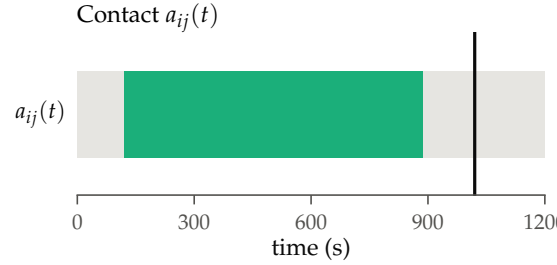
$t = 1020$ s  $j$ passes by $i$: no  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



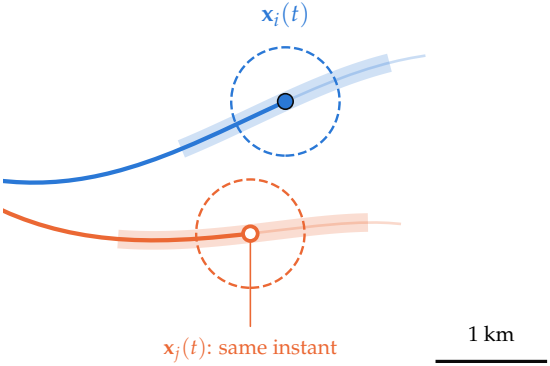



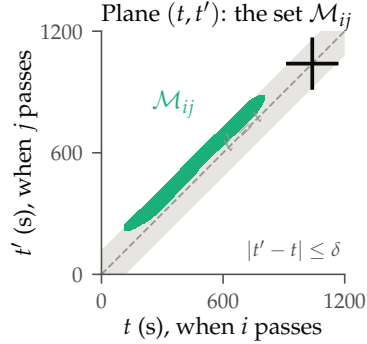



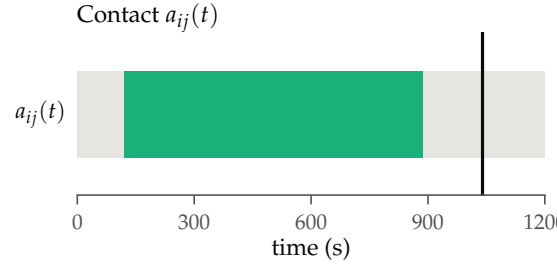
$t = 1040$ s  $j$ passes by $i$: no  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



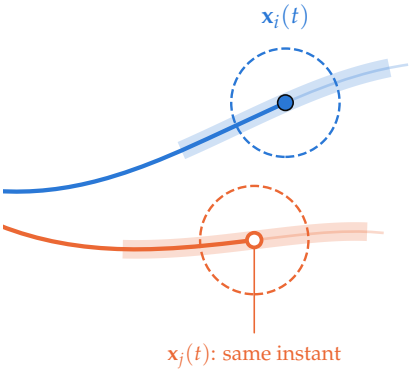



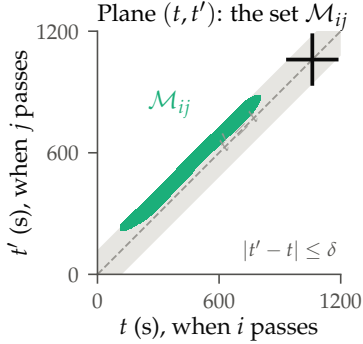
$t = 1060$ s  $j$ passes by $i$: no  $i$ passes by $j$: no

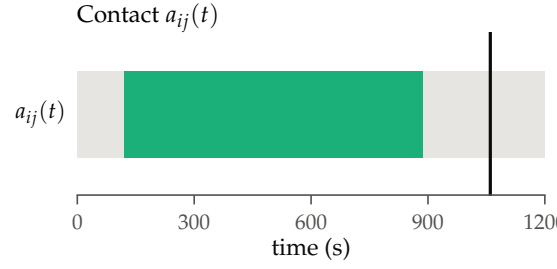
**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$




1 km

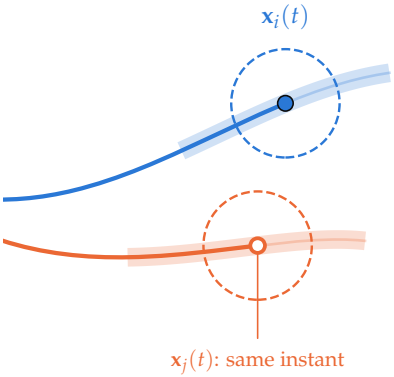


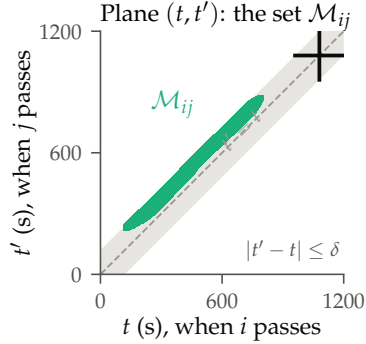
$t = 1080$ s  $j$ passes by $i$: no  $i$ passes by $j$: no

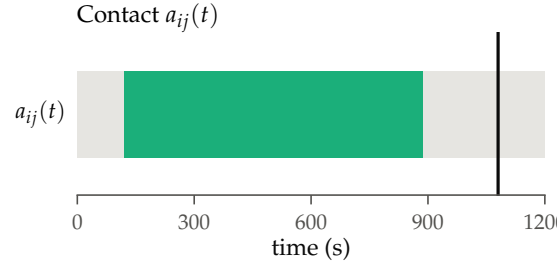
**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$




1 km

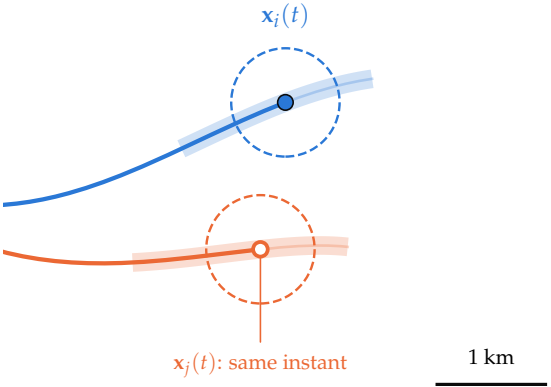


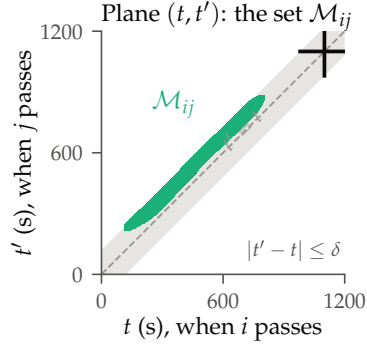



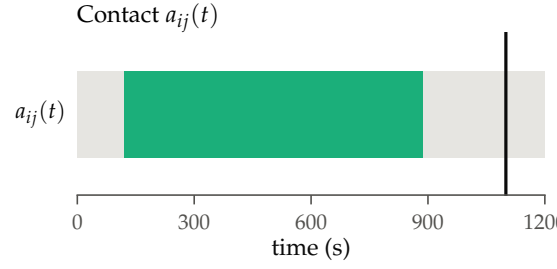
$t = 1100$ s  $j$ passes by $i$: no  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



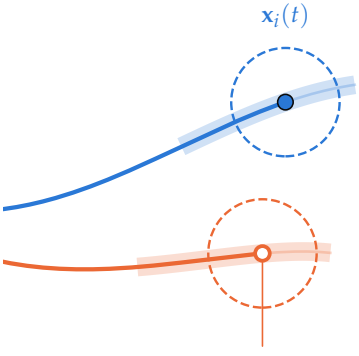



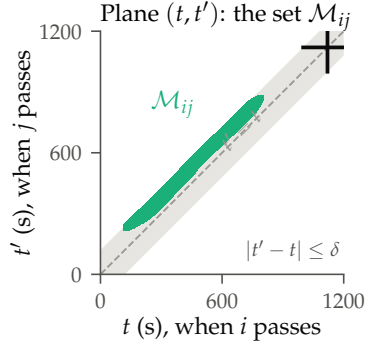
$t = 1120$ s  $j$ passes by $i$: no  $i$ passes by $j$: no

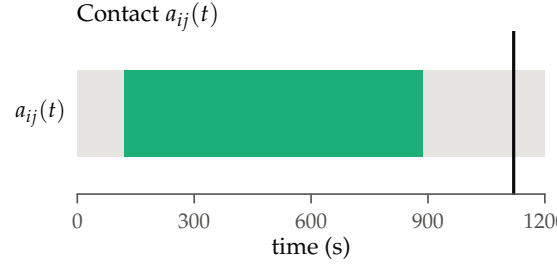
**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$




1 km

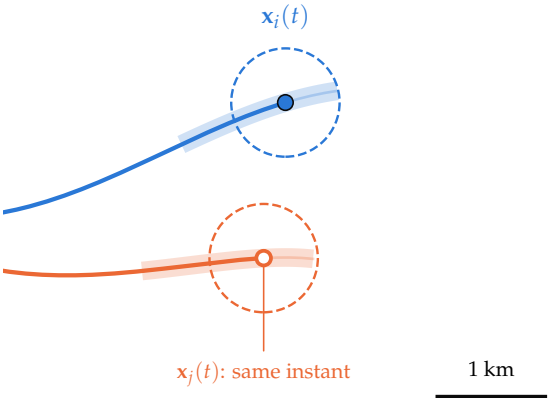


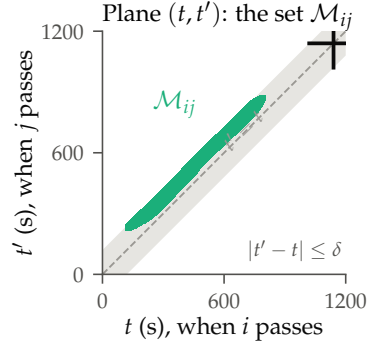



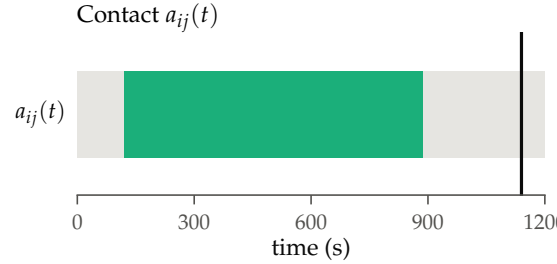
$t = 1140$ s  $j$ passes by $i$: no  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



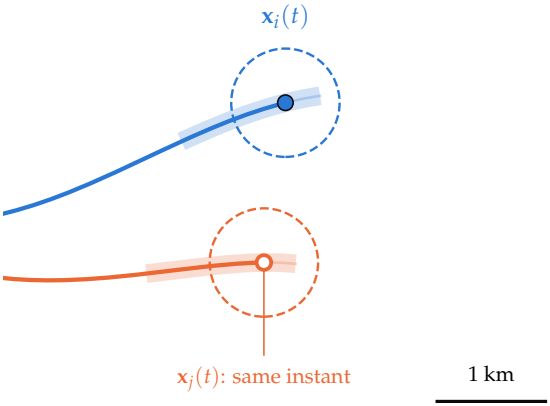



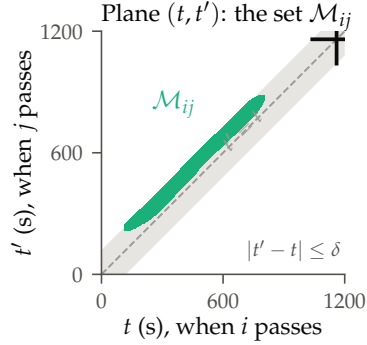



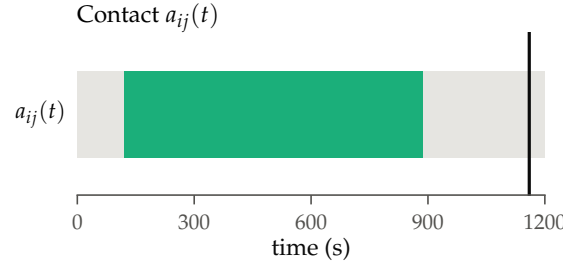
$t = 1160$ s  $j$ passes by $i$: no  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



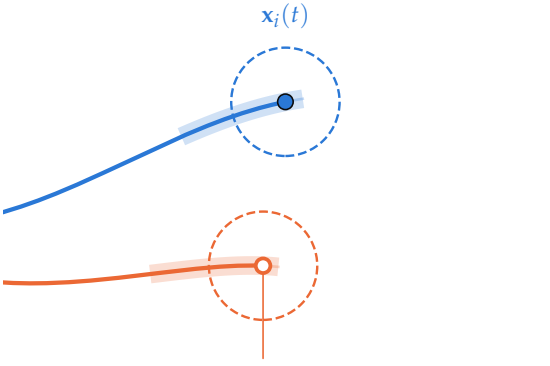



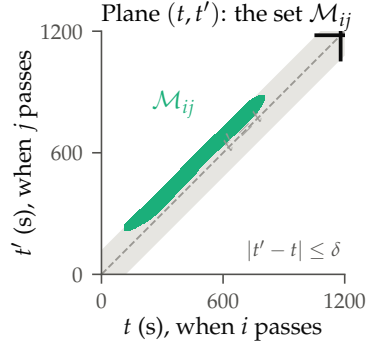



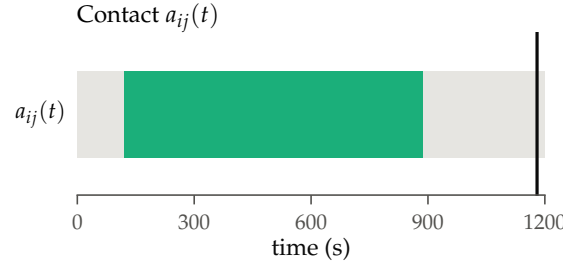
$t = 1180$ s  $j$ passes by $i$: no  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



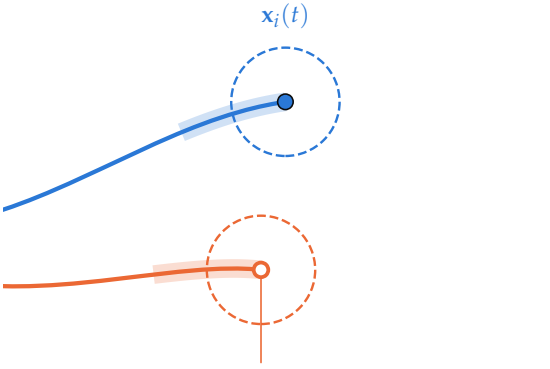



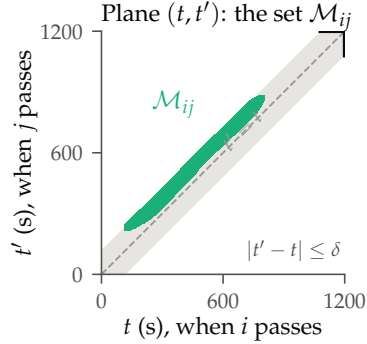



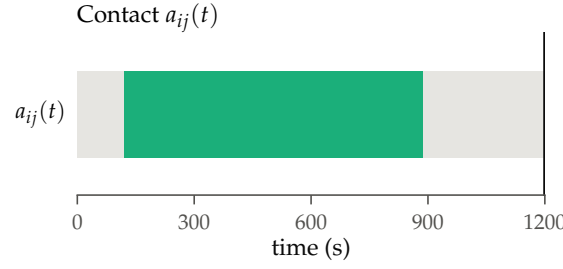
$t = 1200$ s  $j$ passes by $i$: no  $i$ passes by $j$: no

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$



$t = 420$ s  $j$ passes by $i$: yes  $i$ passes by $j$: yes

**Plan view**

circles: radius $\varepsilon_h = 500$ m
bands: each pilot from $t - \delta$ to $t + \delta$

